Project Specification Document

Project: Aegis Analytics AI System

Application Scenarios
A web-based AI analytics assistant for analyzing data, generating insights, dashboards, and reports for students, developers, researchers, and businesses.

Agent Architecture
Frontend (React) → FastAPI Backend → AI/LLM Services → Analytics Engine → Database/Storage.

Core Capabilities
• Data upload and analysis
• AI-generated insights
• Interactive dashboard
• Report generation
• User-friendly web interface

Model & Local Deployment
The application uses a FastAPI backend and can be deployed locally using Python and required dependencies. Environment setup and execution instructions are provided in the GitHub README.

AMD Radeon GPU Optimization
The project is designed to support GPU acceleration where compatible AI frameworks support AMD Radeon GPUs through ROCm. Future optimization includes native ROCm inference for improved performance.

Technology Stack
Python, FastAPI, React, HTML/CSS/JavaScript, REST APIs.

GitHub Repository
https://github.com/Uditrajput09/aegis-analytics-ai-system

Future Enhancements
Multi-model support, advanced visualizations, ROCm-native optimization, and collaborative analytics.